First measurements against real models — V0 and V3c

Runs: `results/v0-20260814-140941/` and `results/v3c-20260814-140941/` · **Backend:** local Ollama, three families (`llama3.2:3b`, `qwen2.5:3b`, `gemma2:2b`), embeddings `nomic-embed-text` · **Corpus:** the 8-prompt smoke set, one seed, temperature 0

> **Provenance, checked before anything below was read.** `harness_validation_only: false` and `embeddings_degraded: false` in both runs — real models, real embeddings, not the mock. `transport_retries: 0` — no connection was dropped and re-tried, so nothing here is a partial result stitched together. τ_sem = 0.51, calibrated on **72 labelled pairs** ($F_{0.5}$ = 0.988, precision 1.00, recall 0.944).
>
> **Reproducibility.** These runs reproduce an earlier pair, cell for cell, at the same seed. The pipeline is deterministic at temperature 0, which is what makes a disagreement between two runs meaningful rather than expected.
>
> **Scale.** Eight prompts, one seed, 2–3B models. A signal to act on, not a benchmark to publish as a headline.

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1. V0 — the coherence tax falls monotonically in ρ

BooookScore-like tax against the monolithic baseline, *k* = 1, 21 valid cells per ρ:

ρ (target)	ρ (achieved)	Coherence tax	Absolute difference	Cells
1.00	1.17	**+24.1 %**	+0.1235	21
1.25	1.27	**+20.4 %**	+0.0761	21
1.50	1.53	**+16.1 %**	+0.0678	21
2.00	2.08	**+13.7 %**	+0.0517	21

Both the ratio and the denominator-free absolute difference decrease with ρ. This is what hypothesis H1 predicts: more shared context per fragment, less quality lost to reassembly. It is the first evidence that the context budget is the variable the design says it is.

The go/no-go criterion is met

Six of 28 category × ρ cells fall below the 5 % threshold fixed before any data existed:

Category	ρ	Tax
creative_writing	2.00	**−9.0 %**
synthetic_data	1.50	**−6.2 %**
synthetic_data	1.25	**−5.1 %**
synthetic_data	2.00	**−0.3 %**
synthetic_data	1.00	**+1.3 %**
code_shared_state	1.50	**+3.2 %**

Negative means fragmentation *improved* the answer on that instrument. `synthetic_data` clears the threshold at every ρ tested. The criterion was written as "at least one task category" because nobody expected all of them to pass — and they do not.

Two measurement failures, reported because they bound the table above

The entity grid is unusable on this corpus. Its monolithic baseline ranges from 0.000 to 0.114 across all eight prompts, median 0.024 — every one below the 0.15 floor at which a ratio stops being a statistic. All **96 of 96** cells are excluded and the harness reports the relative entity-grid tax as *not measured* rather than as zero. An earlier version of this run, before the denominator was checked, reported figures as extreme as **−180 %**.

One prompt broke its baseline. `rag_summarization_filings` produced a monolithic answer of one sentence and six tokens — a generation failure, not a coherence result. Its BooookScore baseline is 0.000 and its 12 cells are excluded. This needs investigating before the next run: a prompt that cannot produce a baseline cannot contribute a tax.

2. V3c — agreement does not predict quality here

Sweeping *k* ∈ {1, 3, 5} at ρ = 1.5, one replica per family:

k	Coherence tax	Mean agreement	HIGH	LOW
1	+13.2 %	—	—	—
3	**+30.8 %**	0.577	29.1 %	42.3 %
5	**+33.3 %**	0.728	58.3 %	28.7 %

Consensus costs roughly **17 to 20 points of quality** relative to *k* = 1. What it buys was supposed to be the confidence map:

> **Pearson r = −0.030 over 597 semantic units.**

The bins are flat and non-monotone, and agreement does not order them: the highest-scoring bin is 0.6–0.8, the *lowest*-agreement bin is second, and the bin where the models agreed most scores below both.

Agreement	Units	Judged acceptable
0.0 – 0.2	40	97.5 %
0.2 – 0.4	91	91.2 %
0.4 – 0.6	80	91.3 %
0.6 – 0.8	122	99.2 %
0.8 – 1.0	264	91.3 %

Section 11.4 of the whitepaper committed in advance: "*a flat or negative correlation would invalidate the confidence map as a reliability signal, and that outcome must be publishable.*" It is published.

The run is not the experiment the whitepaper specifies

The judge accepted 93.3 % of units. With that little variance in the dependent variable a correlation cannot appear even if the underlying signal is real. This measurement therefore **cannot distinguish**:

1. agreement between independent model families does not predict correctness; or
2. a peer-class judge does not discriminate quality finely enough to detect it.

Section 11.4 specifies V3c against **ground-truth datasets**. This run used the judge — the weaker instrument the whitepaper already warns about. **The honest statement is that the confidence map is unsupported, not refuted.**

That does not rescue the claim. An unsupported property cannot be advertised as the architecture's most valuable one, and it no longer is: Section 1.3 was rewritten, Section 8.4b carries the measured result, E16 is now disclosed explicitly as a mechanism with no claimed reliability benefit, and L13 records the whole thing as a limitation.

3. What changed in the paper

Change	Where
Results reported in full, including the unfavourable one	new Section 11.3
Abstract states both outcomes	Abstract
The confidence map demoted from "most valuable property" to "mechanism, benefit unmeasured"	Section 1.3, point 2
The measured correlation and its weakness stated at the mechanism	Section 8.4b, point 2
E16 disclosed as a mechanism with no reliability claim	Section 13
L13 — the confidence map has no demonstrated value	Section 12
L14 — the entity grid does not function on short answers	Section 12
Metrics table gains a *measured* column	Section 11.5
Conclusion revised: first measurement in, one contribution weakened	Section 14

4. What to run next, in priority order

1. **V3c against ground truth.** The single experiment that would settle whether the confidence map is worth its 17–20 point cost. Everything else about E16 is speculation until this exists.
2. **Fix `rag_summarization_filings`.** A prompt that yields a six-token baseline is a bug in the corpus or in generation, not a finding.
3. **Replace or repair the second coherence instrument.** One working mechanical proxy is thinner evidence than this design deserves.
4. **Longer outputs, more prompts, more seeds.** Eight prompts and one seed bound how much any of the above can mean.

5. Reproduce

```
./scripts/run_ollama.sh v0     # results/v0-<stamp>/
./scripts/run_ollama.sh v3c    # results/v3c-<stamp>/
```

Runs from commit `846aeea` onwards record the denominator alongside every ratio, exclude and count unstable cells, take the headline from $k = 1$, report an unmeasured point as absent rather than as zero, and survive a dropped connection.